STAT A411/611 Class Notes

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Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Plan and Chapter 3 Review

Today: Chapter 4, Sections 4.1 and 4.2.
Homework: Chapter 3: 2, 4, 6, 8, 10, 12, 14, 20, 22(a).
Review: Chapter 3 - Discrete random variables and their probability distributions.

Sections 3.1–3.3: Basic Objects

A discrete random variable is described by

$$X, \qquad \mathcal{X}, \qquad \text{pmf}, \qquad \text{cdf}, \qquad E[h(X)].$$

For a function h, the expected value is

$$\boxed{E[h(X)] = \sum_{x \in \mathcal{X}} h(x)p(x)}.$$

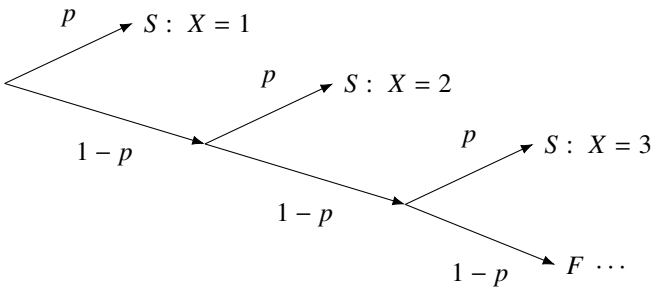
Section 3.4: Discrete Uniform and Geometric Distributions

For k equally likely possible values $x_1, \ldots, x_k$,

$$\boxed{X \sim U(k)}, \qquad P(X = x_i) = \frac{1}{k} \quad (i = 1, \ldots, k).$$

For independent Bernoulli trials with success probability p, if X is the number of trials until the first success,

$$\boxed{X \sim G(p)}.$$



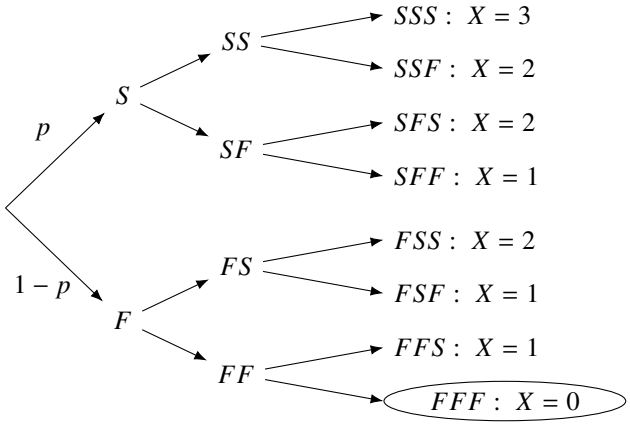
Section 3.5: Binomial Distribution

For n independent Bernoulli trials, let X be the number of successes. Then

$$\boxed{X \sim B(n,p)}, \quad \mathcal{X} = \{0, 1, \ldots, n\},$$

$$\boxed{P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}}, \qquad \boxed{E[X] = np}.$$

For $n = 3$, the eight possible success/failure sequences are



Section 3.6: Negative Binomial Distribution

If X counts trials until the rth success, then

$$\boxed{X \sim NB(r,p)}, \qquad \mathcal{X} = \{r, r+1, r+2, \ldots\},$$

with $NB(1,p) = G(p)$.

Distribution	Trials	Successes	$\mathcal{X}$
$G(p)$	X (random)	1	$\{1, 2, \ldots\}$
$B(n,p)$	n	X (random)	$\{0, 1, \ldots, n\}$
$NB(r,p)$	X (random)	r	$\{r, r+1, \ldots\}$

Section 3.7: Hypergeometric Distribution

From a population of N elements, with M successes, sample n without replacement. Let X be the number of successes in the sample:

$$\boxed{X \sim HG(N,M,n)}, \qquad \boxed{E[X] = n\frac{M}{N}},$$

$$\boxed{P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}}.$$

Here $\binom{N}{n} = \dfrac{N!}{n!(N-n)!}$.

Section 3.8: Poisson Distribution

If $X \sim P(\lambda)$, then its PMF is

$$\boxed{P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}}, \qquad x = 0, 1, 2, \ldots.$$

Counts of earthquakes, accidents, and emergency-room arrivals are possible applications. For a Poisson process with rate α events per unit time, let X be the number of events in an interval of length t:

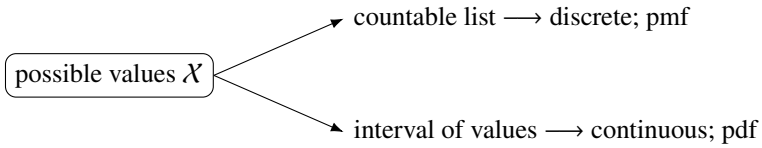
$$\boxed{X \sim P(\alpha t)}.$$

Chapter 4 - Continuous Random Variables and Their Probability Distributions

Section 4.1 - Continuous Random Variables and Probability Density Functions

Discrete versus Continuous Random Variables

A random variable X assigns a number to each experimental outcome. Its support $\mathcal{X}$ is the set of possible values.



The density-based continuous models use intervals such as $[0, 1]$ or $(0, \infty)$.

$$\underbrace{\sum}_{\text{discrete}} \longleftrightarrow \underbrace{\int}_{\text{continuous}}.$$

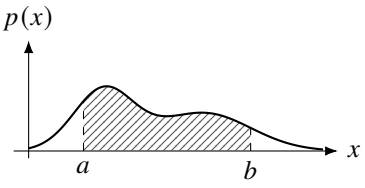
Probability Density Function (pdf)

A valid pdf $p(x)$ satisfies

1. $p(x) \geq 0$ for all $x \in \mathbb{R}$;
2. $\displaystyle\int_{-\infty}^{\infty} p(x)\,dx = 1.$

The probability of an interval is its area under $p(x)$:

$$\boxed{P(a \leq X \leq b) = \int_a^b p(x)\,dx}.$$

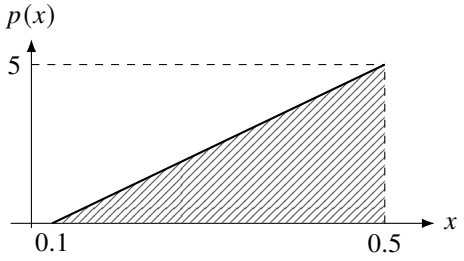


Example 4.1.1: Lead Concentration in Gasoline

Let X be the lead concentration in a liter of gasoline (a nonnegative quantity). Consider the pdf

$$\boxed{p(x) = \begin{cases} 12.5x - 1.25, & 0.1 \le x \le 0.5, \\ 0, & \text{elsewhere.} \end{cases}}$$

For this model $\mathcal{X} = [0.1, 0.5]$.



(a) Is this a valid pdf? The density is nonnegative and the total area is a triangle:

$$\int_{-\infty}^{\infty} p(x)\,dx = \frac{1}{2}(0.5 - 0.1)(5) = \frac{1}{2}(0.4)(5) = \boxed{1}.$$

Hence $p(x)$ is a valid pdf.

(b) Find $P(0.2 \le X \le 0.3)$

$$\begin{aligned} P(0.2 \le X \le 0.3) &= \int_{0.2}^{0.3} (12.5x - 1.25)\,dx \\ &= \left[6.25x^2 - 1.25x\right]_{0.2}^{0.3} \\ &= \left(6.25(0.3)^2 - 1.25(0.3)\right) - \left(6.25(0.2)^2 - 1.25(0.2)\right) \\ &= (-0.1875) - (-0.375) = \boxed{0.1875}. \end{aligned}$$

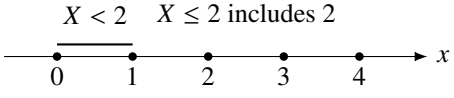
Interval Endpoints

For a continuous random variable, $P(X = a) = 0$; therefore

$$P(X < a) = P(X \le a), \qquad P(a < X < b) = P(a \le X \le b).$$

In contrast, for a discrete variable $X \sim B(20, 0.4)$,

$$P(X < 2) = P(X \le 1) \ne P(X \le 2).$$



Cumulative Distribution Function

For a continuous random variable,

$$\boxed{F(x) = P(X \le x)}, \qquad -\infty < x < \infty.$$

The cdf is nonnegative and nondecreasing:

$$F(x) \ge 0 \quad \forall x, \qquad a \le b \Longrightarrow F(a) \le F(b).$$

(c) Find the cdf of Example 4.1.1

For the lead-concentration density,

$$F(x) = \begin{cases} 0, & x < 0.1, \\ \displaystyle\int_{0.1}^{x} (12.5t - 1.25)\, dt, & 0.1 \le x \le 0.5, \\ 1, & x > 0.5. \end{cases}$$

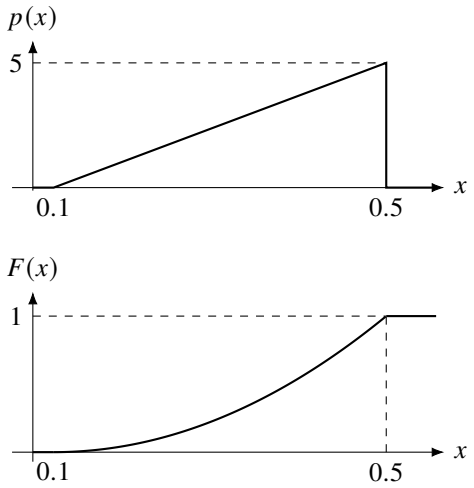
When $0.1 \le x \le 0.5$,

$$\begin{aligned} F(x) &= \int_{-\infty}^{0.1} 0\, dt + \int_{0.1}^{x} (12.5t - 1.25)\, dt \\ &= \left[6.25t^2 - 1.25t\right]_{0.1}^{x} \\ &= 6.25x^2 - 1.25x - \left(6.25(0.1)^2 - 1.25(0.1)\right) \\ &= \boxed{6.25x^2 - 1.25x + 0.0625}. \end{aligned}$$

Hence

$$\boxed{F(x) = \begin{cases} 0, & x < 0.1, \\ 6.25x^2 - 1.25x + 0.0625, & 0.1 \le x \le 0.5, \\ 1, & x > 0.5. \end{cases}}$$

pdf and cdf Graphs



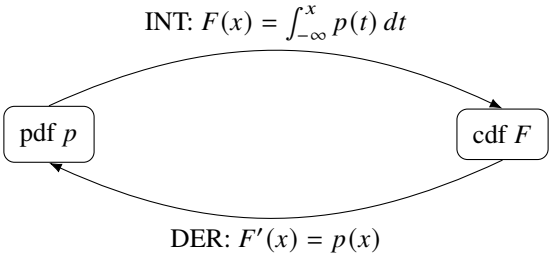
(d) Find $P(0.2 \le X \le 0.3)$ using the cdf

$$\begin{aligned} P(0.2 \le X \le 0.3) &= P(X \le 0.3) - P(X < 0.2) \\ &= P(X \le 0.3) - P(X \le 0.2) \\ &= F(0.3) - F(0.2) \\ &= 0.25 - 0.0625 = \boxed{0.1875}. \end{aligned}$$

Relationship Between the pdf and cdf

$$\boxed{F(x) = \int_{-\infty}^{x} p(t)\,dt}, \qquad \boxed{F'(x) = p(x)}$$

by the Fundamental Theorem of Calculus (where p is continuous).

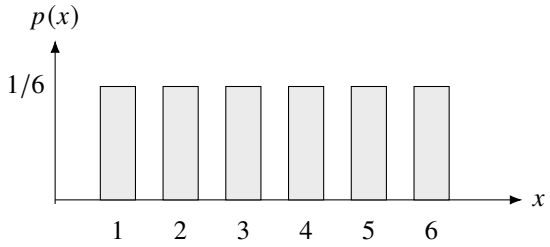


Discrete Uniform Distribution

If X takes k equally likely values $x_1, \ldots, x_k$, then

$$X \sim U(k), \qquad P(X = x_i) = \frac{1}{k}, \quad i = 1, \ldots, k.$$

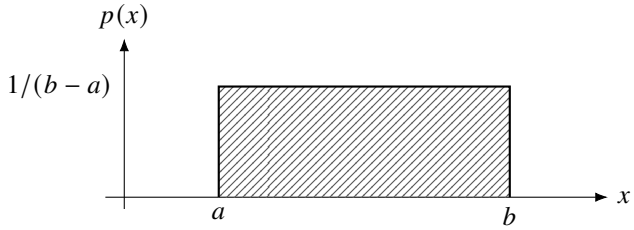
For a fair six-sided die, $X \sim U(6)$ and $P(X = i) = 1/6$ for $i = 1, \ldots, 6$.



Continuous Uniform Distribution

If $X \sim U(a, b)$, where $a < b$, the density is

$$\boxed{p(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{elsewhere.} \end{cases}}$$



The rectangle has width $b - a$ and height $1/(b - a)$, giving area 1.

cdf of a Continuous Uniform Random Variable

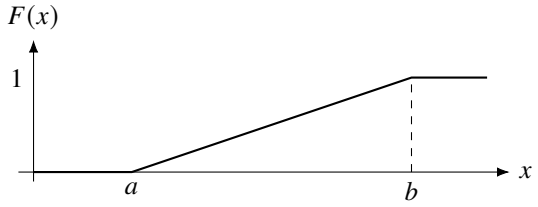
For $X \sim U(a, b)$,

$$\boxed{F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}}$$

For $a \le x \le b$,

$$\begin{aligned} F(x) &= \int_{-\infty}^{a} 0 \, dt + \int_{a}^{x} \frac{1}{b-a} \, dt \\ &= \frac{1}{b-a} [t]_a^x = \frac{x-a}{b-a} \\ &= \frac{1}{b-a} x - \frac{a}{b-a}. \end{aligned}$$

Continuous Uniform cdf Graph



Homework

Exercises 2, 4, 6, 8, 10, 12, 14 (pp. 139–141).

Section 4.2 - Expected Values of Continuous Random Variables

For a discrete random variable (Section 3.3),

$$E[h(X)] = \sum_{x \in \mathcal{X}} h(x)p(x).$$

For a continuous random variable, the corresponding definition is

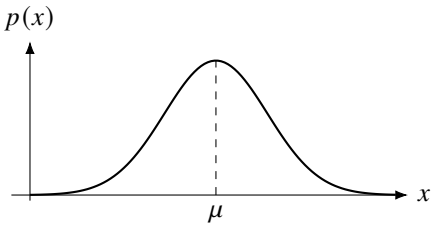
$$\boxed{E[h(X)] = \int_{-\infty}^{\infty} h(x)p(x)\,dx}. \qquad (*)$$

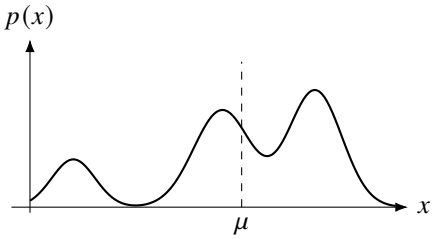
Case I: $h(x) = x$ **- The Mean**

Taking $h(x) = x$,

$$\boxed{\mu = E[X] = \int_{-\infty}^{\infty} x\,p(x)\,dx}.$$

The mean μ measures the center of the distribution.

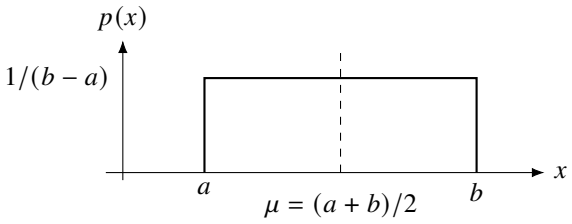




Mean of a Continuous Uniform Random Variable

If $X \sim U(a, b)$, then

$$\boxed{E[X] = \frac{a+b}{2}}.$$



Proof.

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x p(x)\, dx \\ &= \int_{-\infty}^{a} 0\, dx + \int_{a}^{b} \frac{x}{b-a}\, dx + \int_{b}^{\infty} 0\, dx \\ &= \frac{1}{b-a}\left[\frac{1}{2}x^2\right]_a^b \\ &= \frac{b^2 - a^2}{2(b-a)} \\ &= \frac{(b+a)(b-a)}{2(b-a)} = \boxed{\frac{a+b}{2}}. \end{aligned}$$

Case II: $h(x) = ax + b$ **- Linearity of Expectation**

For constants a, b,

$$\boxed{E[aX + b] = aE[X] + b}.$$

Proof.

$$\begin{aligned} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b) p(x)\, dx \\ &= a \int_{-\infty}^{\infty} x p(x)\, dx + b \int_{-\infty}^{\infty} p(x)\, dx \\ &= aE[X] + b \cdot 1 = aE[X] + b. \end{aligned}$$

Hence $E[\cdot]$ is a linear operator.

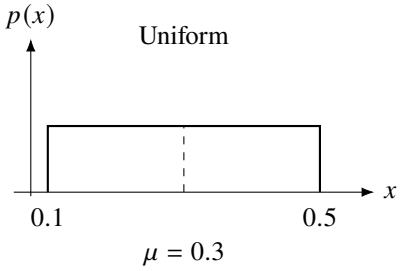
Example 4.2.1: Mean of the Lead-Concentration Model

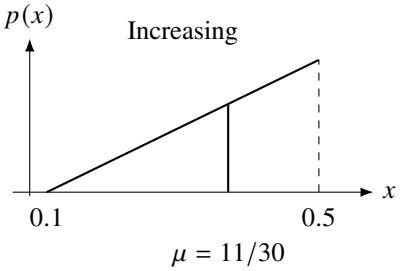
For $p(x) = 12.5x - 1.25$ on $[0.1, 0.5]$ and zero elsewhere,

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x p(x)\, dx \\ &= \int_{-\infty}^{0.1} 0\, dx + \int_{0.1}^{0.5} x(12.5x - 1.25)\, dx + \int_{0.5}^{\infty} 0\, dx \\ &= \int_{0.1}^{0.5} (12.5x^2 - 1.25x)\, dx \\ &= \left[\frac{12.5}{3}x^3 - \frac{1.25}{2}x^2\right]_{0.1}^{0.5} \\ &= \boxed{\frac{11}{30} \approx 0.3667}. \end{aligned}$$

Comparison of the Means

For $X \sim U(0.1, 0.5)$, $E[X] = 0.3$ is the midpoint. The increasing density in Example 4.2.1 has $E[X] = 11/30 \approx 0.3667$, to the right of 0.3.





Case III: $h(x) = x^k$ - Moments

For $k = 1, 2, 3, \ldots$,

$$\boxed{E[X^k] = \int_{-\infty}^{\infty} x^k p(x)\, dx}.$$

This is the kth moment of X. In particular, $E[X]$ is the first moment and $E[X^2]$ the second. In general,

$$\boxed{E[X^2] \neq (E[X])^2}.$$

Case IV: $h(x) = (x - \mu)^2$ **- Variance and Standard Deviation**

With $\mu = E[X]$, the **variance** is

$$\boxed{V[X] = E[(X - \mu)^2] = \sigma^2}.$$

Equivalently,

$$\boxed{V[X] = E[X^2] - (E[X])^2}.$$

The **standard deviation** is

$$\boxed{\text{sd}[X] = \sigma = \sqrt{V[X]}}.$$

Both densities below have mean μ. The narrower density has smaller standard deviation.

